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RESEARCH ARTICLE

Deception Technology Based Intrusion Protection and Detection Mechanism for Digital Substations: A Game Theoretical Approach

DEVIKA JAY^{ID}, (Member, IEEE)

Gridsentry Private Ltd., Bengaluru 560058, India

e-mail: devikajay@gmail.com

ABSTRACT Securing substations from cyber attacks is essential to safeguard critical power infrastructure. However, digital substations that are based on the IEC-61850 standard have Generic Object Oriented Substation Events (GOOSE) messages and Sampled Value (SV) messages that are time-critical and thus cannot be protected using encryption techniques. This work presents a study on deception technology (decoys) for mitigating cyber attacks on GOOSE message virtual LAN (VLAN) which is a non-observable strongly connected biography. In this paper, the deployment of defender decoys is proposed by defining observable subgraphs in the VLAN. The defender-attacker interaction is modeled as a single-leader single-follower game with the defender as the leader. The optimal allocation of decoys for asset protection and attack detection is then formulated as a bi-level optimisation problem. Simultaneous allocation and sequential allocation of protection and detection decoys are considered for defender resource allocation. The existence of equilibrium of the defender-attacker game is proven. The model is illustrated in a 3-IED VLAN and performance is evaluated in a 12-IED VLAN system in the PSRC-I5 protection relay report. The results are compared with the zero-sum game model and it is found that the proposed model is capable of mitigating attacks in the GOOSE VLAN

INDEX TERMS Deception technology, IEC-61850 standard, intrusion protection mechanism, intrusion detection mechanism, single-leader single-follower game.

NOMENCLATURE

VARIABLES

$P(n)$	No. of observable subgraphs.
$D(t)$	Delay in traffic flow.
α	Arrival curve of traffic at a node.
β	Service curve of traffic at node.
T	Latency term in service curve.
σ	Burst parameter of traffic flow.
R	Rate term of service curve.
$\Delta D(t)$	Change in delay caused by cyber attack.
ΔZ	Residual vector due to change in system states.
V_t	Voltage state of the system at time t .
$T_{s,s'}^A$	Probability of attacker to succeed.
$T_{s,s'}^D$	Probability of defender to succeed.
$T_{s,s'}^{PD}$	Probability of protection decoy to succeed.

$T_{s,s'}^{DD}$	Probability of detection decoy to succeed.
ω_1	Integer variable corresponding to residual vector, $\in \{0,1\}$.
ω_{2i}	Integer variable corresponding to change in delay at i^{th} subgraph, $\in \{0,1\}$.
R^A	Expected reward for attacker.
R^D	Expected reward for defender.
R_{sim}^D	Expected reward for defender through simultaneous allocation of decoys.
R_{seq}^D	Expected reward for defender through sequential allocation of decoys.
r_1	final attack residue on delay and voltage.
r_2	intermediate attack residue on delay and voltage after protection decoy deployment.
c_{aj}	Attacker agent deployed at j^{th} subgraph.
d_i	Defender strategy in action space (pd_i, dd_i) .

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- pd_i Protection decoy in defender strategy in j^{th} subgraph.
 dd_i Detection decoy in defender strategy in j^{th} subgraph.
 $C_i = \frac{ca_i}{C^A}$.
 $D_i = \frac{pd_i}{P^D}$.
 $P_i = \frac{dd_i}{D^D}$.

CONSTANTS

- C^A Total number of attacker agents.
 β_j^A Success rate of attacker agent ca_j .
 P^D Total number of protection decoys.
 D^D Total number of detection decoys.
 β_j^{PD} Success rate of protection decoy.
 β_j^{DD} Success rate of detection decoy.

I. INTRODUCTION

With the introduction of IEC-61850 standard, the substation automation system achieved a revolutionary change by moving towards digital substations. Advanced communication and information technologies are used in data acquisition, remote control, monitoring, and operating critical equipment in the digital substation. The interdependency between power and communication network has become very strong, thus making power systems vulnerable to cyber attacks.

In IEC-61850, GOOSE (Generic Object Oriented Substation Events) is an unacknowledged protocol used for inter-IED communication. GOOSE enables the exchange of substation events, status change notifications, alarms, and control commands including circuit breaker status, switch breaker interlock, etc. GOOSE is transmitted in a publish/subscribe fashion to subscriber IEDs over multicast groups.

There are several ways in which an attacker can exploit the vulnerabilities in the network. Attacks are launched by compromising the user interface, station level communication bus, bay level devices, process bus level, interrupting the time synchronization process, and even capturing and modifying GOOSE and SV messages. Another attack strategy that severely affects critical infrastructure is the Advanced Persistent Threats (APT). APTs remain unnoticed in the network until the consequences become visible in the system and the damage has occurred. Such APTs cannot be detected using conventional security tools.

Replay and masquerade attacks can be performed on GOOSE messages that will impair the protection and control functionalities of the substation [1]. The attacker can also flood the network with false GOOSE messages and thereby creating Denial of Service (DoS) attacks. Attackers can also interrupt the GOOSE messages and create malfunctioning of the substation. Tampering of GOOSE messages is also targeted by attackers [2]. The various attack methods like changing critical data frames in a GOOSE message have been detailed in [3] and termed as GOOSE Poisoning.

However, the IEC-61850 standard does not specify security features to address these cyber threats on substation communication network. Thus IEC-62351 standard was published to address this security gap in GOOSE. However, due to the time-critical nature of GOOSE and SV messages, commonly practiced security measures on other domain have limited applications in substations. As per IEC 62351 recommendation, usage of encryption techniques for GOOSE is not recommended. Thus substation communication network security still relies significantly on cyber-hygiene best practices. Several works explored the usage of encryption-based measures for securing GOOSE messages [4], [5], but failed to meet the time-criticality constraints. Pre-shared symmetric keys [1], [6] were found to be satisfying the timing constraints, but this required a secure key management technique. Thus in short, an unencrypted communication channel exists between the substation devices making it vulnerable to attacks.

A network-level system for Efficient Detection For GOOSE-based Poisoning attacks (ED4GAP) was proposed in [7] to monitor the substation's local area network (LAN) through port mirroring on switches. The method also analyzes the GOOSE messages to identify indicators of poisoning attacks. A collaborative intrusion detection mechanism was proposed in [8] as a single IDS may not be effective for an entire substation. This collaboration of IEDs with in-built IDS was suggested through MAC address to secure the substation. The IEDs are to have an intrusion detection indicator, GOOSE and SV attack indicator, and access violation indicator. A multi-dimensional intrusion detection mechanism which includes Access control detection, Protocol whitelisting detection, a Model-based detection approach to analyze Substation Configuration Details (SCD) files and normal IEC 61850 traffic contents, and a Multi-parameter based detection to identify possible threats against SCADA was proposed in [9]. The techniques to defend APT are [10]

- 1) Limited resource based defense strategy like honeypots
- 2) Dynamic information flow tracking (DIFT)
- 3) Cloud based defense strategy for APT attacks on cloud

Specification-based algorithm to detect violation of predefined security rules in multi-cast messages of IEC 61850 was proposed in [11]. Network behaviour-based detection system that can monitor suspicious behaviours when a traffic packet is not following protocol sequence has been proposed in [12]. In [13] a technique to detect GOOSE spoofing attack considering $stNum$, $sqNum$, $timeAllowedtoLive$, T , and payload of the GOOSE dataframe was proposed.

In the above papers on intrusion detection strategies, attacks based on manipulation of the GOOSE data frame were considered. However, these techniques fail to detect advanced persistence threats in the system. Such attacks though difficult to be detected, are persistent, evasive and complex. Thus it is required to include an intrusion protection system in addition to intrusion detection system for GOOSE messages

TABLE 1. Research gap and motivation.

Reference	Intrusion detection	Intrusion protection	Defensive deception with Game theory
[8]	✓	×	×
[1]	×	✓	×
[4]	×	✓	×
[5]	×	✓	×
[6]	×	✓	×
[7]	✓	×	×
[11].	✓	×	×
[12]	✓	×	×
[13]	✓	×	×
Proposed work	✓	✓	✓

to offer a comprehensive cyber security mechanism for digital substations. This is summarised in Table 1

In this paper, a defensive deception technology (decoys) based intrusion protection and detection mechanism is proposed for securing the GOOSE and SV messages in digital substations. The advantage of using deception technology is its capability of handling APT and does not require any encryption or digital signature techniques. Hence it does not affect the time-critical nature of the GOOSE and SV messages. A game theoretical analysis of the intrusion protection mechanism (IPS) and intrusion detection mechanism (IDS) is presented in this work to understand the optimal decoy allocation to perform the intrusion protection and detection mechanism.

The main contributions of the paper are

- 1) A framework for the deployment of decoys for suitable IPS and IDS mechanism
- 2) The optimal decoy allocation for IPS and IDS scheme in the substation communication network by determining the number of observable subgraphs in the Virtual Local Area Network (VLAN) defined in the substation communication network.
- 3) A game theoretical analysis of attacker-defender game as a single leader-single follower game with the proof of existence of equilibrium is presented in the work. The game is then modelled as a bilevel mixed integer nonlinear optimisation problem to obtain the optimal decoy allocation strategy. The defender game model can be utilised as an online learning algorithm for decoys to improve their real-time efficiency.

The paper organisation is as follows: The defensive deception technology is explained in Section II. The decoy allocation in VLAN by identifying observable subgraphs in presented in Section III. The system model, attacker model and defender models are detailed in Section IV. The game theoretical analysis with proof of existence of equilibrium is provided in Section V. Simulation results are discussed in Section VI. Brief conclusions are drawn in Section VII.

II. DEFENSIVE DECEPTION TECHNOLOGY

Deception has been a warfare strategy since ages and has been widely used in military tactics. When the deception strategy is used in the defender mode, it is generally called defensive deception technology. This will prevent the attackers from

TABLE 2. Defensive deception techniques for asset protection and attack detection.

Objective	Technology	Expected impact
Asset protection	Masking	Blending, hiding, misleading
	Decoy	Luring, misleading
	Repackaging	Blending, hiding, misleading
	Dazzling	Confusing, hiding, misleading
	Camouflaging	Blending, hiding, misleading
	False information	Confusing, Luring, misleading
	Lies	Hiding, misleading
Attack detection	Displays	Hiding, misleading
	Mimicking	Luring, misleading
	Inventing	Luring, misleading
	Decoy	Luring, misleading
	False information	Confusing, Luring, misleading

discovering vulnerabilities and also help defenders to study the attacker methods.

The techniques used in defensive deception technology are

- 1) Masking: To hide information in the background
- 2) Repackaging: To hide information as something else
- 3) Dazzling: To hide through overshadowing
- 4) Mimicking: To imitate aspects of real object
- 5) Inventing: To create new fake object
- 6) Decoying: To provide false information available public

The objectives of defensive deception technology is to provide asset protection and attack detection. The defensive deception technologies that provide these objectives are classified and tabulated in Table 2 [14].

The advantage of using decoy-based deception technology is that a wide range of attacker strategies like Scanning attacks (Reconnaissance), Network fingerprinting, Denial of service, Malware, Privacy attacks, Advanced persistent threats (APT), and Masquerades can be handled. For a substation, a decoy is required to hide the actual communication network (VLAN-Virtual LAN) and also mislead the attacker to a false VLAN network. This makes the decoy suitable for protecting substations and detecting attacks even on GOOSE without encryption techniques.

A protection decoy for the protection of GOOSE messages can be developed by mimicking the GOOSE VLAN. An attacker which approaches via the station switch can be lured to the decoy using techniques like port mirroring, open virtual switch, etc. A detection decoy can initiate alarm when such an event occurs in the network. The detection decoy shall also be capable of detecting any changes in the message packets in the VLAN using deep packet inspection techniques. When the attacker is trapped in the protection decoy it then interacts with the decoy. Thus the actual VLAN is hidden from the attacker thereby mitigating cyberattacks on the actual system. The interaction of attacker with the protection decoy can be utilised to generate alarms for the system operator and initiate an incidence response.

A network deception technology called Reconnaissance Deception System based on Software Defined Networking (SDN) to defend reconnaissance was developed in [15]. The deception is achieved by simulating or emulating a virtual network topology. The most commonly used decoy system

is a honeypot which has been explored in smart grid environment in [16]. The work presented the design and implementation of smart grid honeypot system to emulate the entire smart grid communication network including multiple substations. An adaptive honeypots based on Partially Observable MDP (POMDP) was discussed in [17]. An integrated defense technique for Internet-of-Things (IOT) to achieve intrusion prevention using decoy system with moving target defense was presented in [18]. An adaptive deception technology for APT29 and APT3 scenario descriptions in the MITRE ATT&CK was proposed in [19]. A low interaction honeypot that simulated any IEDs compatible with IEC 61850 was proposed in [20]. But the low-interaction honeypot was not capable of handling GOOSE messages.

Game theory is capable of providing a framework that is suitable for defining attacker and defender moves, and payoffs/rewards. Thus game theory is suitable for evaluating the efficiency of implementing the deception technology. Game theoretical analysis is being used in other frameworks like channel access strategy [21] which shows that the game theory is a strong tool for theoretical analysis for strategy definition. Cyber deception games are incomplete and imperfect information games. They can be modeled as partially observable stochastic game [22]. An asymmetric information one-shot Bayesian game was proposed in [23]. In [24] Masking game was introduced to model the attacker-defender games. Other methods of modeling the attacker-defender interaction are dynamic Bayesian games [25] and Hypergame [26], [27].

However, in the above works, an attacker-defender game model for protecting substation Virtual LAN (VLAN) through deception technology was not discussed. The existing models may not be useful in studying the attack-defender game in substations because two strategies (asset protection and attack detection) are to be considered in a VLAN and there are multiple VLANs in the network. The defender aims to protect each VLAN and also the entire substation network in a coordinated manner. Thus it is necessary to study the allocation of decoys in each VLAN for asset protection and attack detection to obtain a secure substation communication network.

Bayesian game models are based on a belief concept about the opponent's motive. Hypergame models are used when the players have a different perception of the game. In this paper, asset protection decoys are deployed in the VLAN for protecting GOOSE messages, and attack detection decoys are deployed in the VLAN for detecting intrusion in GOOSE messages. Thus optimal decoy deployment in the observable subgraphs shall be studied by minimizing the impact of the attack on the network and not based on the belief function. Here the perception of the game by players is based on the rewards issued from the system model that is the same for both attacker and defender. Thus hypergame modeling is also not required in this study.

In this work, an architecture for deploying decoy agents on multiple VLANs in the network is discussed. The optimal

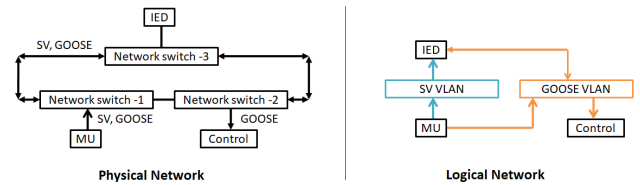


FIGURE 1. Physical and logical network at process bus level.

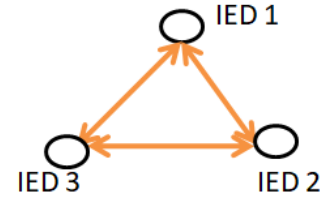


FIGURE 2. Flow diagram of GOOSE message in a VLAN with three IEDs.

allocation of decoys in the VLAN for asset protection and attack detection is then modeled as a defender-attacker game which is presented as a single-leader single follower game with proof of the existence of equilibrium. The model is then formulated as a bilevel optimisation problem for obtaining the optimal decoy allocation strategy in VLAN.

III. DECOY ALLOCATION IN VIRTUAL LANs

In IEC 61850 standard based substation communication network, Virtual LANs (VLANs) are defined in the IEEE 802.1Q standard to allow separation of traffic using logical network definition. By logical separation of the message path using VLAN, multi-cast messages like GOOSE are forwarded only to the designated VLAN ID. Thus GOOSE and SV flow on separate VLAN in the process bus level. In the bay bus level GOOSE VLANs are defined according to the substation configuration. An illustration of the physical network and VLAN logical network at the process bus level is shown in Fig 1.

Let us consider the network switch at the bay/process bus level communication which has three IEDs physically connected to the switch and communicating with GOOSE message through VLAN. The VLAN flow diagram of GOOSE message is as shown in Fig 2. Using graph theory, it can be understood that the GOOSE flow diagram in a VLAN as shown in Fig 2 is a strongly connected bidirectional graph (SCBG). For efficient asset protection and attack detection in a VLAN using decoys, the SCBG should be observable.

The graph observability problem was first studied in [28]. Let $\mathcal{G} = (V, E)$ be a digraph. V is the set of vertices and E stands for a set of edges. A digraph is fully observable if and only if the out-degree of a vertex is not more than 1 [29]. The VLAN graph is a strongly connected bidirectional graph. By graph theoretical principles, a digraph is fully observable if and only if the out-degree of a vertex is not more than 1. Thus a VLAN with 'n' IEDs as vertex, being a strongly connected bidirectional graph, is not observable. This makes VLAN not observable by a single agent like decoy which is

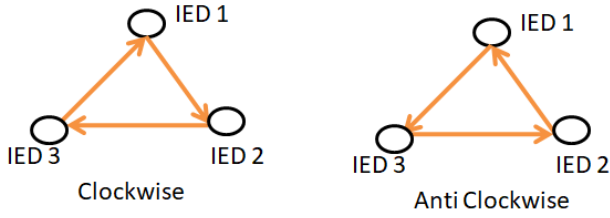


FIGURE 3. Observable ring subgraphs in VLAN.

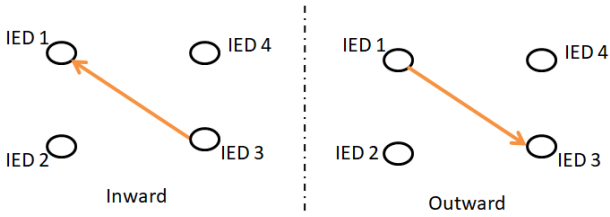


FIGURE 4. Observable subgraphs from a single node.

capable of intrusion an protection of packets flowing in the VLAN. Thus, multiple agents (decoys) are to be deployed in a VLAN to obtain complete observability. The path to be observed by each agent is defined using subgraphs. The subgraphs are defined in such a way that, each subgraph is observable.

The direct and easiest way of defining such subgraphs will be to define subgraphs with two nodes and unidirectional edges. However, if we consider such independent path in the VLAN for deployment of decoy, then for a VLAN with ' n ' IEDs as vertex, the number of decoys required will be $n(n-1)$ which will not be feasible when n is high. Hence subgraphs are to be defined such that the number of decoys to be deployed for complete observability of the graph is less.

Hence, in this work, it is proposed to define observable subgraphs from the VLAN-SCBG, so that the number of decoys required for asset protection and attack detection is reduced. For easy understanding and without loss of generality, consider the case of three IEDs connected in a VLAN as in Fig 2. The two subgraphs that can be defined from the VLAN graph are as shown in Fig 3.

The rings shown in Fig 3 represent the inward and outward cyclic GOOSE messages connecting all IEDs in the VLAN. Let us now consider a single IED node and its inward and outward GOOSE message flow in the network excluding the adjacent nodes in the cycle defined in Fig 3. This is illustrated in Fig 4 with a VLAN having 4 IED nodes. From a single IED vertex, there are $(n-3)$ paths for inward and outward GOOSE message flow. Then the total number of inward and outward message flow subgraphs within a VLAN with ' n ' IEDs considering single node structure is $\frac{n(n-3)}{2}$, if n is even. If n is odd, $\frac{(n+1)(n-3)}{2}$. The inward and outward subgraphs can be combined as a single subgraph assuming that decoys are capable of tracking bi-directional messages. These graphs along with the two inward and outward cyclic graphs defined in Fig 3, together define the observable subgraphs in the strongly connected bigraph VLAN. Thus the total number of

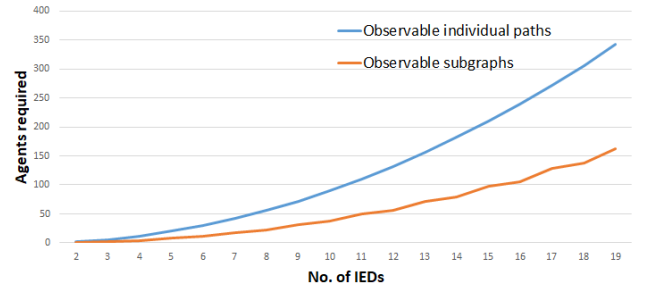


FIGURE 5. Comparing observable individual paths with observable subgraphs as VLAN size increases.

observable subgraphs in a VLAN with ' n ' IEDs is given by

$$P(n) = \begin{cases} 2 + \frac{n(n-3)}{2} & n \text{ is even} \\ 2 + \frac{(n+1)(n-3)}{2} & n \text{ is odd} \end{cases} \quad (1)$$

A detailed explanation for the derivation of the observable subgraphs is provided in Appendix A. The advantage of defining subgraphs instead of individual observable paths is shown in Fig 5. The number of agents required to observe the entire VLAN is significantly reduced when observable subgraphs are defined.

It can be noted that these subgraphs are defined on the same vertex set and hence the sum of the Laplacian of subgraphs will be equal to the Laplacian of the SCBG. This is necessary for preserving the spectral properties of the graph.

The total number of observable subgraphs for a VLAN with ' n ' IEDs is given by $P(n)$ in equation (5) whereas the total number of observable individual paths is given by $n(n-1)$. $P(n)$ is lesser than $n(n-1)$ when $n > 2$. For example when $n = 3$ as in Fig 3, the number of observable individual paths = 6. This implies that 6 decoys are to be deployed to achieve complete observability of the VLAN. Whereas if observable graphs are considered, $P(n) = 2$. Thus only 2 decoys are to be deployed to observe the VLAN. This makes the deployment of decoys in the VLAN to observe the GOOSE messages easier and provides the required IPS and IDS in the system.

IV. GAME MODELING

In this section, the system model and the attacker-defender strategies are detailed. Decoys are proposed in this work for substation VLAN protection and attack detection. Decoys have been widely used in cybersecurity measures for the IT and ICS industry [30], [31]. The decoy system hides the actual VLAN network and misleads the attacker to a false VLAN network. Thereby, protecting the GOOSE messages from attackers and detecting any attack in the network. The defender's objective is to achieve a dynamic optimal protection and detection decoy allocation in the VLAN in response to the attacker's strategies. The targets of the attacker considered in this work are

- 1) To introduce delay in the communication network (Denial of service)

- 2) To change the relay settings/parameters (False trip signal/alarm)

A. SYSTEM MODELING

In this section, the system parameters like communication delay in the VLAN GOOSE messages and relay settings or system parameters in the physical system are discussed to model the cyber-physical system. According to network calculus theory [32], [33], the delay in traffic-flow at time (D(t)) is bounded by

$$D(t) \leq h(\alpha, \beta) = \sup\{\inf[\tau \geq 0; \alpha(s) \leq \beta(s + \tau)]\} \quad (2)$$

where α is the arrival curve and β is the service curve of the traffic at a node. A (σ, ρ) model is generally used as arrival curve model where σ is called the burst parameter and ρ is the upper bound on the long-term average rate of the traffic flow. The service curve model is generally considered as a rate-latency type, $\beta_{R,T}$, where R is the rate term and T is latency term. A linear model is considered for service model, $\beta_{R,T} = R[t-T]^+$. The linear model is widely used for service curve type and is called the rate-latency type. Then the delay is defined as [34],

$$D(t) \leq T + \frac{\sigma}{R} \quad (3)$$

Upon attack, the delay is altered to

$$D(t) \leq \tilde{T} + \frac{\tilde{\sigma}}{\tilde{R}} \quad (4)$$

Thus the change in delay introduced by the attacker is defined as

$$\Delta D(t) = D(t) - \tilde{D}(t) \quad (5)$$

$$\Delta D(t) \leq \Delta T + \frac{\Delta \sigma R - \Delta R \sigma}{R(R + \Delta R)} \quad (6)$$

where $\Delta T = T - \tilde{T}$, $\Delta \sigma = \sigma - \tilde{\sigma}$, $\Delta R = R - \tilde{R}$. When the attacker aims at altering the relay settings or system parameters, the state of the system changes. Let before attack, the system be defined as

$$x_{t+1} = Ax_t + Bu_t + w_t \quad (7)$$

Upon attack of δ_t on the system input, the system state equation changes to

$$x_{t+1} = Ax_t + B(u_t + \delta_t) + w_t \quad (8)$$

Then the residual vector (ΔZ) is defined as

$$\Delta Z = x_{t+1} - x_{t+1} = B(\delta_t) \quad (9)$$

The objective of the attacker is to maximise the change in delay ($\Delta D(t)$) in the VLAN network paths and the residual vector (ΔZ) of the system. The objective of the defender is to minimise $\Delta D(t)$ and ΔZ .

The parameters considered for identifying an attack in the VLAN are ΔD and ΔZ . For incorporating the ΔZ parameter in the bi-level optimisation problem that models the defender-attacker game, it is assumed that the attacker aims at creating

a fault scenario in the substation. Assuming no disturbance in the system, under a no-attack scenario, the voltage state for the next time stamp (V_{t+1}) is equal to the voltage state at the current time stamp (V_t). Upon attack, voltage state for the next time stamp $V_{t+1} = V_t + \delta_t$. Under system disturbance case with no-attack scenario also next time stamp $V_{t+1} = V_t + \Omega_t$, where Ω_t is the noise due to disturbance. In the model presented in the paper, the target of the attacker is to create a false trip scenario which makes $V_{t+1} = 0$. This holds even if the system disturbances are considered. Without loss of generality a no-disturbance, no-attack scenario is considered a normal event in the substation. If the attacker succeeds to create a fault scenario in the substation, then $V_{t+1} = 0$. Then $\delta_t = -V_t$. If the defender succeeds, the fault scenario is averted, and $\delta_t = 0$. Thus the residual vector is defined as

$$\Delta Z = \begin{cases} -V_t & ; \text{If attacker succeeds} \\ 0 & ; \text{If defender succeeds} \end{cases} \quad (10)$$

The probability of attacker to succeed is given by $T_{s,s'}^A$ and the probability of defender to succeed is given by $T_{s,s'}^D$. Let $T_{s,s'}^A - T_{s,s'}^D = \gamma_1$. Then, ΔZ can be re-written as

$$\Delta Z = \begin{cases} -V_t; & \gamma_1 > 0 \\ 0; & \gamma_1 \leq 0 \end{cases} \quad (11)$$

Integer programming is utilised to handle the piece wise function ΔZ . Let $\omega_1 = \{0,1\}$ be an integer variable. If $\gamma_1 > 0$, then $\omega_1 = 1$, else if $\gamma_1 \leq 0$, then $\omega_1 = 0$. Thus linearising the piece wise function ΔZ

$$\Delta Z = -V_t \cdot \omega_1 \quad (12)$$

$$-(1 - \omega_1) < \gamma_1 \leq \omega_1 \quad (13)$$

Thus if attacker succeeds, i.e. $\gamma_1 > 0$ implies that $\omega_1 = 1$ and $\Delta Z = -V_t$. If defender succeeds, $\omega_1 = 0$ and $\Delta Z = 0$.

Similarly, a delay is introduced in the i^{th} observable subgraph if the attacker's success on the i^{th} subgraph is greater than the success probability of the defender on the i^{th} observable subgraph. Let γ_{2i} be defined as the difference of attacker success on the i^{th} observable subgraph and success probability of defender on the i^{th} observable subgraph.

Then, the delay introduced in i^{th} observable subgraph of the VLAN by attacker is ΔD_i is defined as,

$$\Delta D_i = \begin{cases} 1; & \gamma_{2i} > 0 \\ 0; & \gamma_{2i} \leq 0 \end{cases} \quad (14)$$

To linearise, define $\omega_{2i} = \{0,1\}$ an integer variable such that, if $\gamma_{2i} > 0$, then $\omega_{2i} = 1$, else if $\gamma_{2i} \leq 0$, then $\omega_{2i} = 0$. Thus,

$$\Delta D_i = \omega_{2i} \quad (15)$$

$$-(1 - \omega_{2i}) < \gamma_{2i} \leq \omega_{2i} \quad (16)$$

Consider a game model $\langle S, A, D, R^A, R^D \rangle$, where S is the state space, A is the action space of the attacker, D is the action space of the defender. R^A is the expected reward of the attacker and R^D is the expected reward of the defender. T_{s^i, s^j} is the transition probability from state s^i to s^j due to attacker action $a \in A$ and defender action $d \in D$.

B. ATTACKER MODELING

The attacker strategy is to allocate attacker agents across the VLAN observable subgraphs $P(n)$. Let the action space of the attacker, $A = \{a_1, a_2, \dots, a_{NA}\}$, such that $a_i = [ca_1^i, ca_2^i, \dots, ca_{P(n)}^i]$ such that $\sum_{j=1}^{P(n)} ca_j^i = C^A$, where C^A is the total agents available with the attacker and ca_j is the agent deployed at j^{th} observable subgraphs. Then the expected reward of the attacker, R^A for attacker action $a \in A$ and defender action $d \in D$ is given by

$$R^A = \sum_{s' \in S} T_{s,s'}^A [\Delta Z + \sum_{i=1}^{P(n)} \Delta D_i] \quad (17)$$

where $T_{s,s'}^A$ is the probability of failing the system by maximising the instantaneous reward $[\Delta Z + \sum_{i=1}^{P(n)} \Delta D_i]$. The probability of failing the system is the probability of success for the attacker. The objective of the attacker is to maximise R^A . Let β_i^A be the success rate of resource ca_i . Then

$$T_{s,s'}^A = \sum_{i=1}^{P(n)} \frac{ca_i}{C^A} \left(\sum_{j=1}^{P(n)} \beta_j^A ca_j \right) \quad (18)$$

The success probability in deploying ca_i at i^{th} observable subgraphs depends on the deployment of attacker agents in other subgraphs. Thus the probability for ca_i at i^{th} observable subgraph is considered as the ratio with respect to the total success rate of all attacker agents.

C. DEFENDER MODELING

The defender strategy is to allocate intrusion protection and intrusion detection decoys for the observable subgraphs in the VLAN. Let the action space of the defender, $D = \{d_1, d_2, \dots, d_{NA}\}$, such that $d_i = (pd_i, dd_i)$, where pd_i is the IPS decoy allocation and dd_i is IDS decoy allocation. The IPS decoy allocation $pd_i = [pd_1^i, pd_2^i, \dots, pd_{P(n)}^i]$ such that $\sum_{j=1}^{P(n)} pd_j^i = P^D$, where P^D is the total decoys available with the defender for IPS and pd_j is the decoy deployed at j^{th} observable subgraph. The IDS decoy allocation $dd_i = [dd_1^i, dd_2^i, \dots, dd_{P(n)}^i]$ such that $\sum_{j=1}^{P(n)} dd_j^i = D^D$, where D^D is the total decoys available with the defender for IDS and dd_j is the decoy deployed at j^{th} observable subgraph. Then the expected reward of the defender, R^D for attacker action $a \in A$ and defender action $d \in D$ is given by

$$R^D = \sum_{s' \in S} T_{s,s'}^D [\Delta Z + \sum_{i=1}^{P(n)} \Delta D_i] \quad (19)$$

$T_{s,s'}^D$ is the probability of securing the system by minimising the instantaneous reward $[\Delta Z + \sum_{i=1}^{P(n)} \Delta D_i]$. The probability of securing the system is the probability of success for the defender. The objective of the defender is to minimise R^D . Let β_i^{PD} be the success rate of decoy pd_i for IPS system and β_i^{DD} be the success rate of decoy dd_i for IDS system. Then the success probability of the defender is defined as

$$T_{s,s'}^D = T_{s,s'}^{PD} + T_{s,s'}^{DD} \quad (20)$$

where $T_{s,s'}^{PD}$ is the success probability of IPS system and $T_{s,s'}^{DD}$ is the success probability of IDS system. Similar to attack success probability, the IPS success probability and IDS success probability is defined as

$$T_{s,s'}^{PD} = \sum_{i=1}^{P(n)} \frac{pd_i}{P^D} \left(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j \right) \quad (21)$$

$$T_{s,s'}^{DD} = \sum_{i=1}^{P(n)} \frac{dd_i}{D^D} \left(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j \right) \quad (22)$$

There are two strategies in which the defender can deploy the decoy system for IPS and IDS.

- 1) Simultaneous allocation of IPS and IDS decoys
- 2) Sequential allocation of IPS and IDS decoys

In the first strategy, the IPS and IDS decoys are allocated simultaneously and thus the decoy agents receive the same instantaneous reward from the system, i.e., $[\Delta Z + \sum_{i=1}^{P(n)} \Delta D_i]$. Whereas in the second strategy, the IDS system is initiated only when the IPS system received a positive instantaneous reward. Hence this allocation is considered as sequential allocation.

For simultaneous allocation of IPS and IDS decoys, the expected reward of the defender, R_{sim}^D for attacker action $a \in A$ and defender action $d \in D$ is given by

$$R_{sim}^D = \left(\sum_{i=1}^{P(n)} \frac{pd_i}{P^D} \left(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j \right) + \sum_{i=1}^{P(n)} \frac{dd_i}{D^D} \left(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j \right) \right) \cdot [\Delta Z + \sum_{i=1}^{P(n)} \Delta D_i] \quad (23)$$

For sequential allocation of IPS and IDS decoys, the expected reward of the defender R_{seq}^D is defined as

$$R_{seq}^D = \sum_{i=1}^{P(n)} \frac{pd_i}{P^D} \left(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j \right) [\Delta Z' + \sum_{i=1}^{P(n)} \Delta D_i'] + \sum_{i=1}^{P(n)} \frac{dd_i}{D^D} \left(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j \right) \cdot [\Delta Z + \sum_{i=1}^{P(n)} \Delta D_i] \quad (24)$$

where, $[\Delta Z' + \sum_{i=1}^{P(n)} \Delta D_i']$ is the intermediate reward received upon deploying only the IPS decoys. The final reward upon deploying both IPS and IDS determines the reward for the attacker and net impact of the attacker-defender actions in the system, which is $[\Delta Z + \sum_{i=1}^{P(n)} \Delta D_i]$.

V. GAME THEORETICAL ANALYSIS

In the previous section, the system, attacker, and defender strategies in the game models were detailed. In this section, the interaction between the attacker and defender is discussed. The defender-attacker game is modeled as a single-leader single-follower game. Since the intrusion protection mechanism is deployed using defensive decoys,

the defender is treated as a leader in this game model. The defender-attacker game is mathematically formulated as a bi-level optimisation problem by incorporating the system model with suitable approximations as detailed in this section

A. MATHEMATICAL FORMULATION

The defender-attacker single leader single-follower game is modeled as a mixed integer bi-level optimisation problem. When the defender allocates IPS and IDS simultaneously the problem formulation is as given below

1) UPPER LEVEL

$$\min \sum_{i=1}^{P(n)} [P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j) + D_i(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j)].r_1 \quad (25)$$

$$\text{subject to: } \sum_{i=1}^{P(n)} pd_i = P^D \quad (26)$$

$$\sum_{i=1}^{P(n)} dd_i = D^D \quad (27)$$

$$\gamma_1 = \sum_{i=1}^{P(n)} [C_i(\sum_{j=1}^{P(n)} \beta_j^A ca_j) - P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j) - D_i(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j)] \quad (28)$$

$$\gamma_{2i} = C_i(\sum_{j=1}^{P(n)} \beta_j^A ca_j) - P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j) - D_i(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j) \quad (29)$$

$$r_1 = -\Delta Z + \sum_{i=1}^{P(n)} \Delta D_i \quad (30)$$

and Equations (12, 13, 15, 16). and subject to

2) LOWER LEVEL

$$\max \sum_{i=1}^{P(n)} C_i(\sum_{j=1}^{P(n)} \beta_j^A ca_j)r_1 \quad (31)$$

$$\text{subject to: } \sum_{i=1}^{P(n)} ca_i = C^A \quad (32)$$

When the defender strategy is to have sequential IPS and IDS allocation, then the IPS system receives an intermediate reward, $\Delta Z'$ and $\Delta D'_i$. The probability of attacker to succeed is given by $T_{s,s'}^A$ and the probability of IPS to succeed is given

by $T_{s,s'}^{PD}$. Let $T_{s,s'}^A - T_{s,s'}^{PD} = \gamma_3$.

$$\gamma_3 = \sum_{i=1}^{P(n)} [C_i(\sum_{j=1}^{P(n)} \beta_j^A ca_j) - P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j)] \quad (33)$$

Then, the intermediate residual vector reward $\Delta Z'$ can be rewritten as

$$\Delta Z' = \begin{cases} -V_t; & \gamma_3 > 0 \\ 0; & \gamma_3 \leq 0 \end{cases} \quad (34)$$

Let $\omega_3 = \{0,1\}$ be an integer variable. If $\gamma_3 > 0$, then $\omega_3 = 1$, else if $\gamma_3 \leq 0$, then $\omega_3 = 0$. Thus linearising the piece wise function $\Delta Z'$

$$\Delta Z' = -V_t \cdot \omega_3 \quad (35)$$

$$-(1 - \omega_3) < \gamma_3 \leq \omega_3 \quad (36)$$

Similarly, a delay is introduced in the i^{th} observable subgraph if the attacker's success on the i^{th} observable subgraph is greater than the success probability of IPS on the i^{th} observable subgraph. Let γ_{4i} be defined as

$$\gamma_{4i} = C_i(\sum_{j=1}^{P(n)} \beta_j^A ca_j) - P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j) \quad (37)$$

Then, the delay introduced in i^{th} observable subgraph of the VLAN by attacker is $\Delta D'_i$ is defined as the intermediate reward for IPS system,

$$\Delta D'_i = \begin{cases} 1; & \gamma_{4i} > 0 \\ 0; & \gamma_{4i} \leq 0 \end{cases} \quad (38)$$

To linearise, define $\omega_{4i} = \{0,1\}$ an integer variable such that, if $\gamma_{4i} > 0$, then $\omega_{4i} = 1$, else if $\gamma_{4i} \leq 0$, then $\omega_{4i} = 0$. Thus,

$$\Delta D'_i = \omega_{4i} \quad (39)$$

$$-(1 - \omega_{4i}) < \gamma_{4i} \leq \omega_{4i} \quad (40)$$

Then, the mixed-integer bi-level optimisation problem when the defender strategy is to sequentially deploy IPS and IDS decoys is as below:

3) UPPER LEVEL

$$\min \sum_{i=1}^{P(n)} [P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j)r_2 + D_i(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j).r_1] \quad (41)$$

subject to: Equations(12, 13, 15, 16, 26, 27, 28, 29, 33, 35, 36, 37, 39, 40)

4) LOWER LEVEL

Equations (31, 32)

B. EXISTENCE OF EQUILIBRIUM

The defender-attack game is modeled as a single-leader single-follower game, in which optimal allocation of decoy to all observable subgraphs in the VLAN for intrusion protection and detection is considered as the defender objective. Similarly, the optimal allocation of attack agents to all observable subgraphs in the VLAN is the objective of the attacker.

The paper presents another notion of game representation to model the above-mentioned defender-attack game, i.e., multi-leader multi-follower game (MLMF). In this modeling, a leader-follower game is considered for each observable subgraph, which makes the entire defender-attack game for the VLAN a multi-leader multi-follower game.

For simultaneous allocation strategy, the utility/reward function of the leader, i.e., defender in Equation (25) can be re-written as:

$$R^D = \sum_{i=1}^{P(n)} \mathcal{R}_i \quad (42)$$

where

$$\mathcal{R}_i = P_i \left(\sum_{j=1}^{P(n)} \beta_j^{PD} p_{dj} \right) + D_i \left(\sum_{j=1}^{P(n)} \beta_j^{DD} d_{dj} \right) \cdot [\Delta D_i - \Delta Z] \quad (43)$$

Optimal value of R^D is thus

$$R_D^* = \inf \sum_{i=1}^{P(n)} \mathcal{R}_i \quad (44)$$

$$\sum_{i=1}^{P(n)} \inf \mathcal{R}_i \leq R_D^* \quad (45)$$

$\mathcal{R}_i$ can be considered as the reward function of individual decoy on i^{th} observable subgraph, that interacts with the attacker agent on the same path. Thus the single leader single-follower game is modeled as a multi-leaders multi-followers game (MLMF) with shared constraints.

Theorem 1: Consider an MLMF game with potential function π and the set of fixed points in the feasible region non-empty. Then equilibrium for the MLMF game exists if the minimiser of a potential game defined by potential function π exist. The minimiser of potential game exists if

- 1) The potential function π is coercive or
- 2) The constraint space is compact

Proof: To prove the existence of equilibrium in MLMF games with shared constraints, the existence of coercive potential function π is to be proven. Thus the minimiser for potential game exists and by the definition of potential function an equilibrium for the MLMF game also thus exists [35]. The potential function definition is given as in

Definition 1: Given a function $f(x_i, x_{-i})$, then $\pi(x_i, x_{-i})$ is defined as a potential function for a function $f(x_i, x_{-i})$,

$\{x_i, x_{-i}\} \in X$, if for all (x_i, x_{-i}) , the condition

$$\nabla \pi(x_i, x_{-i}) - \nabla f(x_i, x_{-i}) = 0 \quad (46)$$

where, ∇ is $\frac{\partial}{\partial x_i}$, is satisfied

In the defender-attacker MLMF game, the candidate potential function $\pi(p_{di}, d_{di}, \sum_{j \neq i} p_{dj}, \sum_{j \neq i} d_{dj})$ is defined as

$$\begin{aligned} \pi = & \rho_0 + \rho_{1p} p_{di} + \rho_{1d} d_{di} + \rho_{2p} p_{di}^2 + \rho_{2d} d_{di}^2 \\ & + \rho_{3p} p_{di} \sum_{j \neq i} p_{dj} + \rho_{3d} d_{di} \sum_{j \neq i} d_{dj} \end{aligned} \quad (47)$$

When the defender strategy is simultaneous, comparing the gradients of the potential function π and the reward function $\mathcal{R}_i$, the potential function can be written as

$$\begin{aligned} \pi = & \frac{\beta_i^{PD}}{PD} \cdot R \cdot p_{di}^2 + \frac{\beta_i^{DD}}{DD} \cdot R \cdot d_{di}^2 + \frac{\sum \beta_i^{PD}}{PD} \cdot R \cdot p_{di} \sum_{j \neq i} p_{dj} \\ & + \frac{\sum \beta_i^{DD}}{DD} \cdot R \cdot d_{di} \sum_{j \neq i} d_{dj} \end{aligned} \quad (48)$$

where $R = \Delta D_i - \Delta Z$.

When the defender strategy is sequential allocation, the potential function becomes

$$\begin{aligned} \pi = & \frac{\beta_i^{PD}}{PD} \cdot R' \cdot p_{di}^2 + \frac{\beta_i^{DD}}{DD} \cdot R \cdot d_{di}^2 + \frac{\sum \beta_i^{PD}}{PD} \cdot R' \cdot p_{di} \sum_{j \neq i} p_{dj} \\ & + \frac{\sum \beta_i^{DD}}{DD} \cdot R \cdot d_{di} \sum_{j \neq i} d_{dj} \end{aligned} \quad (49)$$

where $R' = \Delta D'_i - \Delta Z'$.

Using Taylor series expansion of the terms $(p_{di} \sum_{j \neq i} p_{dj})$ and $(d_{di} \sum_{j \neq i} d_{dj})$ around the point $(P^D, 0)$ and $(D^D, 0)$ respectively, the potential function in both strategies becomes coercive. Hence the equilibrium point for the MLMF game exists.

VI. SIMULATION RESULTS

In this section, the numerical illustration of decoy-based IPS and IDS in VLAN is presented for a 3-IED VLAN system. The performance of simultaneous and sequential strategies of the defender mechanism in deploying and IDS is evaluated in PSRC-I5 Schematic Representation of Power System Relaying which has 12-IED VLAN system [36]. The results are compared with a zero-sum game approach which represents the state-of-the-art techniques in intrusion protection and detection methods.

A. 3 IEDs CONNECTED IN A GOOSE-VLAN

Consider three IEDs connected in a VLAN for GOOSE messaging as shown in Fig 1. The number of observable subgraphs, $P(n) = 2$ in this case. Only the two rings as shown in Fig 3 are observable subgraphs. Hence the decoy IPS and IDS are deployed in these two observable subgraphs by the

TABLE 3. Sequential allocation of IPS, IDS decoys: 3 IED system subgraph 1; $\beta_2^A = 0.20$.

Attacker β_1^A	IPS decoy allocation (%)	IDS decoy allocation (%)	Attacker agent (%)
0.1	50	50	50
0.3	50	50	50
0.5	50	50	50
0.7	50	50	50
0.9	73.3	0	0

TABLE 4. Sequential allocation of IPS, IDS decoys: 3 IED system subgraph 2; $\beta_2^A = 0.20$.

Attacker β_1^A	IPS decoy allocation (%)	IDS decoy allocation (%)	Attacker agent (%)
0.1	50	50	50
0.3	50	50	50
0.5	50	50	50
0.7	50	50	50
0.9	26.7	100	100

defender. The attacker also deploys its agents in these two observable subgraphs.

First, the case in which the attacker strategy is to attack subgraph 1 with varying probability and attack subgraph 2 with probability, $\beta_2^A = 0.20$ is illustrated. The value for β_i^{DD} and $\beta_i^{PD} \forall i = 0.50$ for the defender agents are considered. When the defender exercises sequential IPS-IDS allocation strategy, the resource allocation in path-1 is as tabulated in Table 3 and that in path-2 is tabulated in Table 4

From Table 3 and Table 4, it can be understood that when the probability of the attacker succeeding in path-1 is below 0.7 the attacker allocates an equal percentage of agents to attack the observable paths. Hence the IPS and IDS system is also allocated in equal percentages. When the attacker probability is higher on subgraph-1, i.e. $\beta_1^A = 0.9$., more IPS decoys are allocated, i.e. 73.3% decoys are allocated for IPS in subgraph-1. This averts the attacker from subgraph-1. The attacker then allocates agents in subgraph-2. However, the remaining IPS decoys and the 100% IDS decoys allocated in subgraph-2 are capable of mitigating the attack. Thus the defender succeeds in the game.

Consider the case when the attacker success on subgraph-2 is higher, that i.e. $\beta_2^A = 0.6$. The sequential IPS-IDS allocation strategy, the resource allocation in subgraph-1 and subgraph-2 is given in Table 5 and Table 6

In Table 5 and Table 6, when the probability of the attacker to succeed in subgraph-1 is 0.5, i.e. almost equal to the success probability in subgraph-2, the attacker allocates all agents on subgraph-1. The attack is completely mitigated by the 80% IPS decoy and 100% IDS decoys allocation in subgraph-1 to mitigate the attack. When the attacker's success in subgraph-1 increases further, the attacker allocates 74.2% agents in subgraph-1 and 25.8% in subgraph-2. 100% IPS

TABLE 5. Sequential allocation of IPS, IDS decoys: 3 IED system subgraph 1; $\beta_2^A = 0.60$.

Attacker β_1^A	IPS decoy allocation (%)	IDS decoy allocation (%)	Attacker agent (%)
0.1	50	50	50
0.3	50	50	50
0.5	80	100	100
0.7	100	0	74.2
0.9	100	0	0.633

TABLE 6. Sequential allocation of IPS, IDS decoys: 3 IED system subgraph 2; $\beta_2^A = 0.60$.

Attacker β_1^A	IPS decoy allocation (%)	IDS decoy allocation (%)	Attacker agent (%)
0.1	50	50	50
0.3	50	50	50
0.5	20	0	0
0.7	0	100	25.8
0.9	0	100	36.7

TABLE 7. Simultaneous allocation of IPS, IDS decoys in 3 IED system in Path 1; $\beta_2^A = 0.60$.

Attacker β_1^A	IPS decoy allocation (%)	IDS decoy allocation (%)	Attacker agent (%)
0.1	52	31	9
0.3	100	48	100
0.5	50	38	45
0.7	100	100	100
0.9	100	100	100

decoys are in subgraph-1 and 100% IDS in subgraph-2 to mitigate the attack.

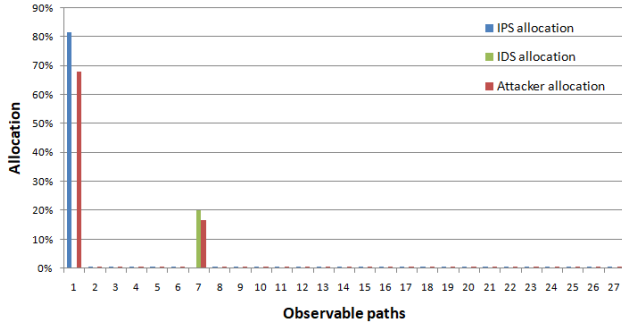
Now, consider the case when the defender strategy is to allocate IPS and IDS decoys simultaneously for the case with the attacker success on path-2, i.e. $\beta_2^A = 0.6$. The allocation in subgraph-1 is tabulated in Table 7

When a simultaneous allocation is done by a defender, the defender wins the game in the cases tabulated in Table 7 and is capable of mitigating attacks. When comparing Table 5 and Table 7, it can be understood that more agents are utilised to mitigate attacks in a single subgraph. Whereas when the strategy of the defender is to allocate IPS and IDS sequentially, the attacker agents are not being concentrated at a single path and the IPS and IDS together in both paths mitigate the attacks. The net effect of attack mitigation on the system through either strategy is the same.

These results are compared with the case when the defender-attacker game is modelled as a zero sum game. The results of the zero-sum game of defender-attacker game in the case when $\beta_2^A = 0.6$ is tabulated in Table 8. When evaluating Table 8, it is understood that the defender mechanism fails when $\beta_1^A = 0.7$ and $\beta_1^A = 0.9$. Hence the sequential/simultaneous IPS and IDS allocation is better when

TABLE 8. Zero-sum game with defender agent allocation in path 1 and 2; $\beta_2^A = 0.60$.

Attacker β_1^A	Defender Path:1(%)	Attacker Path:1(%)	Defender Path:2(%)	Attacker Path:2(%)
0.1	50	100	50	100
0.3	60	100	40	100
0.5	100	100	0	0
0.7	100	0	0	100
0.9	100	0	0	100

**FIGURE 6.** Sequential allocation of IPS and IDS in 12-node VLAN: $\beta_i^A = 0.7 \forall i$.

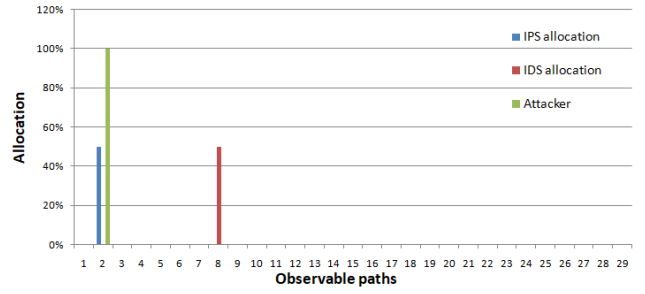
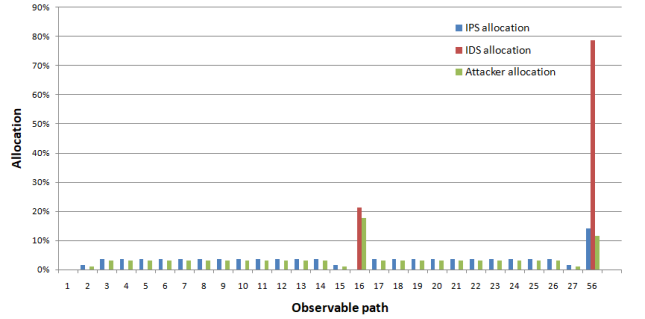
compared to a single defender agent strategy based on zero-sum games.

B. PSRC-IS SCHEMATIC REPRESENTATION OF POWER SYSTEM RELAYING

In the relay communication example cited in the report for Relaying Practices detailed in PSRC-IS Schematic Representation of Power System Relaying, 12 relays- 2 Transformer protection relay, 2- Line protection relays, 2- Bus protection relays, and 6 other relays (for line reactors, etc.)- are connected to a network switch. This network switch will thus correspond to one diagonal in a breaker bus system. For a 12-node VLAN system, the observable subgraphs ($P(n)$) is 56. If individual paths in the strongly connected bigraph were considered for deployment, the number of agents required would have been equal to 132. By considering observable subgraphs for the deployment of agents, the number of agents required is much lesser. In all cases of study, β_i^{PD} and $\beta_i^{DD} = 0.5 \forall i$.

Consider the case of sequential allotment of IPS and IDS decoys in the 12-node system. The IPS-IDS allocation when $\beta_i^A = 0.7 \forall i$, with the attacker allocation is shown in Fig 6.

The same attacker strategy is simulated considering zero-sum game model. The results are shown in Fig 7. It can be observed from Fig 6 that when all nodes have equal attacker success probability, the attacker randomly chooses subgraphs for attacks. In this case, subgraph-1 and subgraph-7 are chosen. In subgraph 1, the attacker deploys 68% of agents which are mitigated by 82% of IPS decoys. 17% agents are deployed by the attacker at subgraph-7 which is mitigated by 20% of IDS decoys. Thereby completing mitigating the attacks and thus defender succeeds.

**FIGURE 7.** Zero-sum game model for IPS and IDS in 12-node VLAN: $\beta_i^A = 0.7 \forall i$.**FIGURE 8.** Sequential allocation in 12-node VLAN: $\beta_{16}^A = \beta_{56}^A = 0.80$.**TABLE 9.** Simultaneous allocation of decoys in 12-node VLAN.

Scenario	Attacker probability	Attacker allocation %	IPS allocation %	IDS allocation %
Attack on all paths	0.9	Path 1:100	Path 1: 94	Path 1: 93
Attack only on selected paths	0.9 (Path 2 and Path 5)	Path 2: 50 Path 5: 50	Path 2: 17 Path 5: 52	Path 2: 28 Path 5: 14
Attack with higher probability on selected paths	0.9 (Path 2 and Path 5); the rest with 0.50	Path 2: 45 Path 5: 55	Path 2: 50 Path 5: 49	Path 2: 61 Path 5: 26

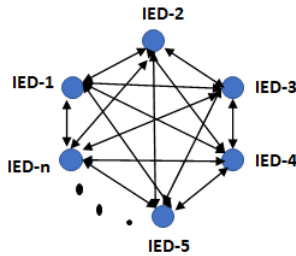
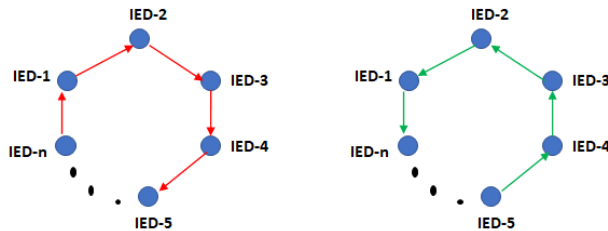
A case in which $\beta_i^A = 0.6$ except for $\beta_{16}^A = \beta_{56}^A = 0.80$ is considered. The resource allocation in this case is shown in Fig 8.

The resource allocation gets distributed throughout the observable subgraphs. Path 16 and path 56 are allocated more agents by the attacker, 18% and 12% respectively. In subgraph 56, IPS mitigates the attack with 14% IPS decoys and 79% IDS decoys. In subgraph 16 the attack is mitigated using 21% IDS decoys.

The simulation results for simultaneous allocation of decoys in 12-node VLAN is tabulated in Table 9. When the probability of success is equal in all paths, the attacker chooses one path and allocated 100% attacker agents. The IPS and IDS together with 94% and 93% decoy allocation respectively mitigate the attack. When the attack is concentrated on path 2 and path 5 with 0.9 probability, the IPS and IDS allocation is distributed in both paths to mitigate the attack. This is shown in case 2 and case 3 in Table 9 with different attacker success probabilities in other paths.

TABLE 10. Attacker-defender game utility functions in 12-node VLAN.

Scenario	Attacker probability	(Attacker,defender) Utility Function	
		Simultaneous allocation	Sequential allocation
Attack on all paths	0.9	(0, 0.225)	(0, 0.5)
Attack only on selected paths	0.9 (Path 2 and Path 5)	(0, 0)	(0, 0)
Attack with higher probability on selected paths	0.9 (Path 2 and Path 5); the rest with 0.50	(0, 0)	(0, 0)

**FIGURE 9. 'n' IEDs in a VLAN as strongly connected bigraph.****FIGURE 10. Two cyclic observable subgraphs.**

The utility value of attacker and defender in simultaneous and sequential allocation strategy is presented in Table 10. The results indicate that when the attack probability is higher, the defender received a higher utility function which implies that the defender strategy is successful in defending the VLAN from attacks. Thus the proposed game theoretical model for decoy allocation in IPS and IDS is capable of mitigating the attacks. This is achieved by suitably defining observable subgraphs in the network which is lesser than the total number of message paths in the network. The allocation of decoys is more effective when IPS and IDS decoys are deployed sequentially. In the sequential allocation strategy for the defender, when IPS decoys are allocated to more sensitive paths, the attackers allocate attacker agents to other paths which are then mitigated by the IDS system. When a simultaneous allocation is done the IPS and IDS decoys together are involved in mitigating the attacks. In both cases, the defender succeeds and thereby protects the substation GOOSE VLAN from attacks.

VII. CONCLUSION

A game theoretical approach to modeling the deployment of intrusion protection and intrusion detection mechanism

is presented in this paper. The technology used for IPS and IDS is deception technology, in which decoys will observe the GOOSE virtual LAN. The defender-attacker interaction is modeled as a single-leader single-follower game in which the defender is considered a leader. However, deploying decoys for all paths in the GOOSE message VLAN is not possible when the number of IEDs is high in the VLAN. Moreover, the VLAN is a strongly connected bigraph that is not observable. Hence in this work, observable subgraphs are defined for deploying decoys for IPS and IDS in the GOOSE VLAN and also to achieve full observability of the VLAN. The advantage of defining subgraphs is that the number of decoys required for the protection and detection of attacks is lesser than the number of individual paths in the VLAN. This makes the deployment of deception technology in GOOSE VLAN easier.

Based on the number of observable subgraphs, the optimal resource allocation of defender agents, i.e., IPS decoys and IDS decoys, is formulated as a bi-level optimisation problem to model the single leader single follower game between the attacker and defender. Two strategies are considered for the defender- simultaneous allocation of IPS and IDS decoys, and sequential allocation of IPS-IDS decoys. These two strategies are found to be effective in mitigating the attacks and achieving optimal allocation of decoys in the observable subgraphs. We are working towards the deployment of decoys in GOOSE VLAN and testing various attack conditions in a substation test bed as the future scope of this paper. The work can be extended to SV VLAN also.

APPENDIX A

OBSERVABLE SUBGRAPHS

Consider 'n' IEDs connected in a VLAN as a strongly connected bigraph (SCBG) as shown in Fig 9. From the SCBG shown in Fig 9, two rings can be defined as shown in Fig 10. One ring represents the inward GOOSE messages and the other ring represents the outward GOOSE messages. Thus we get two observable subgraphs that can be derived from the SCBG.

Further analysing Fig 9, consider a single IED node and its inward and outward GOOSE message flow in the network excluding the adjacent nodes in the cycle defined in Fig 10. This is illustrated in Fig 11. There are (n-3) paths for inward and outward GOOSE message flow, from a single node when the subgraphs are represented as in Fig 11.

Then the total number of inward and outward message flow subgraphs within a VLAN with 'n' IEDs considering single node structure is $\frac{n(n-3)}{2}$, if n is even. If n is odd, $\frac{(n+1)(n-3)}{2}$.

The inward and outward subgraphs can be combined as a single subgraph assuming that decoys are capable of tracking bi-directional messages. These graphs along with the two inward and outward cyclic graphs defined in Fig 10, together define the observable subgraphs in the strongly connected

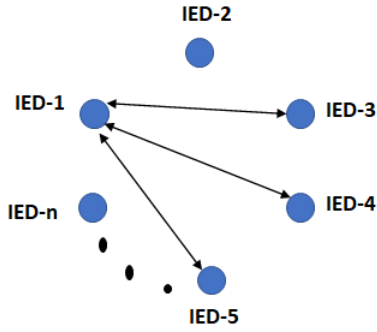


FIGURE 11. Observable subgraphs from a single node excluding adjacent nodes.

bigraph VLAN. Thus the total number of observable subgraphs in a VLAN with 'n' IEDs is given by

$$P(n) = \begin{cases} 2 + \frac{n(n-3)}{2} & n \text{ is even} \\ 2 + \frac{n+1(n-3)}{2} & n \text{ is odd} \end{cases} \quad (\text{A.1})$$

APPENDIX B DEFENDER-ATTACKER MODEL

The objective function of defender and attacker models are explained in this appendix.

Under simultaneous allocation of protection and detection decoy, the defender objective is to minimise

$$\min \sum_{i=1}^{P(n)} [P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j) + D_i(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j)] \cdot r_1 \quad (\text{B.1})$$

The reward is distributed to each observable subgraph based on the success probability of the protection decoy and that of the detection decoy to be deployed in the observable subgraph. r_1 is the reward signal received from the system which comprises of the delay in the VLAN and also the change in voltage in the network. The reward is given by

$$r_1 = -\Delta Z + \sum_{i=1}^{P(n)} \Delta D_i \quad (\text{B.2})$$

where ΔZ is given by

$$\Delta Z = \begin{cases} -V_i; & \gamma_1 > 0 \\ 0; & \gamma_1 \leq 0 \end{cases} \quad (\text{B.3})$$

$\gamma_1 > 0$ implies that the attacker has succeeded in launching an attack leading to trip scenario. This implies that the success probability of attacker is greater than that of the protection and detection decoy on all observable subgraphs which is given by

$$\gamma_1 = \sum_{i=1}^{P(n)} [C_i(\sum_{j=1}^{P(n)} \beta_j^A ca_j) - P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j) - D_i(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j)] \quad (\text{B.4})$$

The delay in an observable subgraph is considered as a binary signal such as

$$\Delta D_i = \begin{cases} 1; & \gamma_{2i} > 0 \\ 0; & \gamma_{2i} \leq 0 \end{cases} \quad (\text{B.5})$$

where delay in observable subgraph is 1 when $\gamma_{2i} > 0$, which implies the success probability of the attacker resources in i^{th} observable subgraph is greater than that of the protection and detection decoys in that subgraph. This is given by

$$\gamma_{2i} = C_i(\sum_{j=1}^{P(n)} \beta_j^A ca_j) - P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j) - D_i(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j) \quad (\text{B.6})$$

Under sequential allocation, defender allocates protection decoys first and then detection decoys. This is because the attackers that are capable of bypassing the protection decoys are still detected by detection decoys to alert the system operator for an attack. Thus the reward for the protection decoy is different from that for the detection decoy. The objective of the defender is to minimise the delay and change in voltage.

$$\min \sum_{i=1}^{P(n)} [P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j) r_2 + D_i(\sum_{j=1}^{P(n)} \beta_j^{DD} dd_j) \cdot r_1] \quad (\text{B.7})$$

The intermediate reward is given by r_2 which is distributed to each observable subgraph based on the success probability of the protection decoy. The intermediate reward r_2 is given by

$$r_2 = -\Delta Z' + \sum_{i=1}^{P(n)} \Delta D'_i \quad (\text{B.8})$$

where $\Delta Z'$ is given by

$$\Delta Z' = \begin{cases} -V_i; & \gamma_3 > 0 \\ 0; & \gamma_3 \leq 0 \end{cases} \quad (\text{B.9})$$

$\gamma_3 > 0$ implies that the success probability of attacker resource is greater than that of the success probability of the protection decoys in all observable subgraphs. Thus γ_3 is defined as

$$\gamma_3 = \sum_{i=1}^{P(n)} [C_i(\sum_{j=1}^{P(n)} \beta_j^A ca_j) - P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j)] \quad (\text{B.10})$$

The intermediate delay signal in i^{th} observable subgraph is given by

$$\Delta D'_i = \begin{cases} 1; & \gamma_{4i} > 0 \\ 0; & \gamma_{4i} \leq 0 \end{cases} \quad (\text{B.11})$$

where $\gamma_{4i} > 0$ implies success probability of the attacker resource in i^{th} observable subgraph is greater than the success probability of the protection decoy in that subgraph.

$$\gamma_{4i} = C_i(\sum_{j=1}^{P(n)} \beta_j^A ca_j) - P_i(\sum_{j=1}^{P(n)} \beta_j^{PD} pd_j) \quad (\text{B.12})$$

The reward for the detection decoy is the final change in the voltage and delay that the attacker could cause in the network which is same as r_1 . The attacker allocates resources to maximise

$$\max \sum_{i=1}^{P(n)} C_i \left(\sum_{j=1}^{P(n)} \beta_j^A c_{aj} \right) r_1 \quad (\text{B.13})$$

where r_1 is the reward observed in the system and the attack resources are allocated in each observable subgraph based on the attacker's success probability.

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DEVIKA JAY (Member, IEEE) received the B.Tech. degree in electrical and electronics engineering from the TKM College of Engineering, Kerala, India, in 2009, and the M.S. degree in power systems and the Ph.D. degree from the Indian Institute of Technology Madras, in 2013 and 2021, respectively. She was the Deputy Manager with Power Grid Corporation of India Ltd. She is currently the Founder and the CTO with Gridsentry Private Ltd. Her research interests include powergrid cybersecurity, power system economics, electricity markets, the application of game theory, and artificial intelligence in power system operation.

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